

Economic Justification and ROI in NHS Commissioning

Digital mental-health and addiction tools are typically justified on a *per-user cost* and *return-on-investment (ROI)* basis. Commissioners compare the low marginal cost of a digital programme to the high costs of traditional care (e.g. therapy sessions or inpatient rehab). For example, NHS analyses found **Sleepio** (a CBT-i app) at £45/year per user is *cost-saving* versus usual care by reducing sleeping-pills and GP appointments ¹. **Wysa** (an AI mental-health chatbot) estimates an annual cost of just £5.90 per eligible user, with modelling showing potential savings by averting ~441,000 waiting-list referrals (each triage costing ~£105) ². **Kooth** (a youth mental-health platform) was modelled to save ~£236 per user per year (via 5,346 fewer GP visits, 298 fewer antidepressant prescriptions and fewer hospitalisations for self-harm) ³. Similarly, a NIHR-funded trial of the **Drink Less** alcohol-reduction app reported reduced consumption and projected “hundreds of millions of pounds” in NHS savings over 20 years if widely scaled ⁴. These examples illustrate that **economic benchmarks** for pilots typically include very low cost-per-user (often <£50/year) and ROI factors of 2–5× (through reduced staff time, medication, or acute episodes) ² ⁴. In designing a pilot, OTOS should plan to capture such metrics – e.g. cost per user, GP/therapy time saved, or avoided hospitalisations – as key outcomes.

Evaluation Frameworks and Evidence Standards

NHS commissioners require digital tools to meet national standards. **NICE's Digital Technology Assessment Criteria (DTAC)** is mandatory: any supplier must demonstrate compliance with safety, data protection, cybersecurity, interoperability, usability and accessibility criteria ⁵. DTAC is the baseline “entry ticket” for procurement. In parallel, the **NICE Evidence Standards Framework (ESF)** defines the required level of clinical evidence by risk/function. A self-guided or coach-guided app typically falls under ESF Tier C or D, meaning it must show credible effectiveness (often via an RCT or robust real-world study) and economic value ⁶. For mental-health/adult ADHD tools, NICE would expect validated outcomes (e.g. PHQ-9/GAD-7 for anxiety/depression, AUDIT/DAST for addiction, or ADHD symptom scales) and a favorable cost-effectiveness profile. The recent NHS ADHD Taskforce explicitly urged that “ADHD services need to be digitised” and recommended NICE “*prioritise rapid assessment of digital products*” ⁷. This means OTOS should align its pilot with NICE ESF levels and be prepared to submit evidence in formats like NICE Technology Adoption Briefings.

Other national frameworks also apply. NHS England and NIHR favour *health-economic evaluations* in pilots (e.g. cost-utility, cost-benefit analyses) ⁸. Where possible, align pilot metrics with NIHR/PH metrics (e.g. quality-adjusted life years, service utilization) and existing NHS KPIs (waiting times, recovery rates). For example, IAPT (Improving Access to Psychological Therapies) services routinely track recovery ($\geq 50\%$ symptom reduction) and waiting time targets; OTOS pilots could similarly benchmark against IAPT performance. In summary, a funding-ready pilot for Cambridgeshire & Peterborough ICS must **build in NICE/NIHR standards**: use validated clinical scales, conduct a health economic analysis, and ensure DTAC compliance (particularly for ADHD/addiction data governance).

Procurement Pathways (Pilot & Rollout)

There are multiple entry routes from pilot to procurement: - **NHS Innovation Accelerator (NIA)** and national programs: Successful innovations often join the NIA to gain NHS-wide visibility. For example, Wysa entered NIA in 2023, and its published ROI case was likely instrumental ². Being on NIA or similar accelerators (e.g. NHS AI Lab collaboratives, NHS Innovation Service) can jump-start adoption.

- **Academic/NIHR pilots:** Partnering with NIHR/Universities can fund rigorous evaluation. The Drink Less app was backed by NIHR and published in *The Lancet*, which builds confidence. Even if NIHR studies don't guarantee commissioning, they generate the evidence NHS trusts look for.

- **ICS and Trust procurements:** Ultimately, regional Integrated Care Boards (ICBs) or ICS digital teams will commission the tool. This often happens via local innovation or transformation budgets. For instance, Sleepio's uptake grew first via early adopters (e.g. Frimley and BOB ICB in 2024) after NICE endorsement ⁹. Suppliers should engage procurement leads or innovation managers in the ICS. Note that NHS contracts for digital products are often placed through standard frameworks: **G-Cloud (Digital Marketplace)** is commonly used for software-as-a-service, allowing ICSs to "call off" contracts quickly ¹⁰. As ICS procurement data show, digital/IT solutions are frequently procured this way. OTOS should consider getting listed on G-Cloud or other NHS frameworks so that post-pilot scaling can transition to standard purchasing.

- **Regional Innovation Hubs and AHSNs:** Health Tech hubs (often at Academic Health Science Networks or Innovation Hubs) connect startups with the NHS. Engaging local hubs (e.g. Health Innovation East in the East of England) can open doors to pilot opportunities and ICS contacts. NIHR-funded programs (e.g. NIHR i4i, Invention for Innovation) or NHS Innovation grants may be options if OTOS seeks research funding that dovetails with NHS pilots.

In summary, OTOS's roadmap should span **pilot funding (research/innovation grants, NIHR, or ICS transformation money)** to **procurement readiness (G-Cloud, ICS contracts, NIA membership)**. Given the NHS emphasis on *value-based procurement*, OTOS must prepare to demonstrate both clinical impact and financial value ¹¹. SMEs should also note the current NHS push for value-based evaluation (new guidance launching in 2026 ¹¹), meaning life-cycle cost savings matter as much as upfront fees.

Case Studies of NHS Digital Tools

- **Sleepio (Big Health)** – a digital CBT program for insomnia. Founded 2010, Sleepio amassed evidence via multiple RCTs ¹². By 2022 NICE recommended it as cost-saving over sleep hygiene ¹³ ¹. Sleepio now has formal NHS funding; e.g. Scottish Govt. funded Sleepio nationally in 2021 ¹⁴, and by mid-2024 it was commissioned in English ICBs (Frimley, BOB) ⁹. Evaluation data include improved Insomnia Severity Index scores and reduced GP prescriptions ¹. Sleepio's case shows the timeline: build evidence (trials, NICE guidance) then engage early adopter regions.
- **Wysa** – an AI chatbot for anxiety/depression. Entered NHS Innovation Accelerator 2023. A real-world study cited by NHS NIA highlighted its low cost (£5.90 per user) and projected reduction of waiting-list backlogs ². Its business case emphasizes therapist-time saved (by handling mild/moderate cases digitally) and redirecting resources to severe cases. Wysa illustrates using NIA data to convince commissioners: low unit cost and service capacity gains.
- **SilverCloud** – an online CBT platform (now part of Amwell). It is used across many IAPT services (Guardian notes ~700k patients referred to SilverCloud's depression/anxiety courses over a decade ¹⁵). SilverCloud clinical trials and economic models (outside the UK) report ~90% lower

delivery cost vs face-to-face ¹⁶. In Wales, SilverCloud is *endorsed free* via local health boards ¹⁷; in England it has no formal NICE MTG but operates widely in IAPT. Its key metrics: high completion rates and time saved per therapist (therapists report ~1.5 hours vs 4 hours per patient) ¹⁵.

- **Drink Less** – an alcohol reduction app. A NIHR-funded RCT (n=5,602) showed modest reduction in drinking (39 units/week vs control) and, importantly, modelled large NHS savings if scaled ⁴. Although not yet in NHS service, this exemplifies how pilot evidence (strong trial data, health economics) can support commissioning (in fact, NICE recommends digital alcohol interventions as add-ons) ⁴.
- **Kooth (Papyrus/Qwell for CYP)** – a free online mental-health and wellbeing service for young people, funded by charities and local authorities. An economic model estimated Kooth delivers ~£236 saving per user (mainly via reduced GP visits and medication) ³. Kooth is currently commissioned by many UK localities. Its lesson is that demonstrating system-wide savings (GP time, reduced acute events) can justify funding by NHS and local government.

These cases share common themes: **low per-user cost** vs high-cost alternatives, **validated clinical outcomes**, and early collection of real-world utilization data. For OTOS, analogous data should be planned: e.g. measure engagement rates, symptom change, and extrapolate service impact (fewer therapy sessions, less crisis care). For context, note that **inpatient addiction rehab** can cost on the order of £14,000 for a 28-day course (~£495/day) ¹⁸, while many digital solutions cost <£100 per user. Framing OTOS as a preventive/step-2 option to reduce expensive interventions will align with these benchmarks.

Designing a Funding-Ready Pilot for Cambridgeshire & Peterborough ICS

To secure ICS support, OTOS's pilot should be tightly aligned with local needs and national standards: - **Partner with Stakeholders:** Engage CPICS mental-health and addiction leads, plus local ADHD specialists, to co-design the pilot. Early liaison ensures the pilot addresses ICS priorities (e.g. reducing wait times for ADHD assessment or reducing alcohol-related harm). Work with CPICS digital or transformation teams to access any existing funding pots for innovation.

- **Define Outcomes Clearly:** Choose validated measures relevant to ADHD and addiction (e.g. ADHD Rating Scale, AUDIT for alcohol, PHQ-9/GAD-7 for comorbidity). Include service metrics (e.g. number of people referred to specialist services, GP appointments, medication use) so ICS can see impact on pathways. Set a clear timeframe (12–18 months) with interim milestones. For example: pilot 200 patients for 6–12 months, measure symptom changes and service utilization before/after.

- **Economic Modelling:** Incorporate an economic evaluation from the start. Project costs and savings under best-case and conservative scenarios (like Wysa and Kooth did) ² ³. Use CPICS data: e.g. *Cambridge* has ~5,775 adults needing alcohol treatment (4,505 untreated) ¹⁹ – use this to model population impact. If OTOS can prevent even a fraction of these cases from needing intensive treatment, the ROI could be substantial. Similarly, tie ADHD intervention to potential avoidance of costly outcomes (e.g. substance misuse, unemployment) – cite the Taskforce's estimate of £17bn avoidable ADHD-related costs ²⁰.

- **NICE/DTAC Preparation:** Make sure the pilot platform complies with NHS digital standards (DTAC). Before or during pilot, conduct a DTAC self-assessment (or have a third party do so) to catch any gaps. Ensure data security and interoperability with NHS systems (e.g. GP records or IAPT systems) to ease integration. Plan to collect evidence on effectiveness and decide which ESF tier applies (likely Tier C for a

hybrid coached intervention). If possible, design the study to meet ESF evidence requirements (controlled comparator or strong pre-post design).

- **Implementation Plan:** Outline how OTOS would be offered (e.g. via GP referral, IAPT, or addiction services). Leverage local digital pathways: for instance, Cambridgeshire's Digital Strategy aims for more online access and shared records ²¹, so show how OTOS integrates. Provide training for clinicians/coaches who will support users.

Benchmarking Outcomes and Economic Metrics

For ICS buy-in, OTOS should propose target **performance metrics**. These might include: - **Clinical Outcomes:** e.g. $\geq 30\%$ reduction in ADHD and addiction symptom scores; proportion achieving "recovery" thresholds (akin to IAPT's $\geq 50\%$ symptom reduction).

- **Engagement:** e.g. $\geq 50\%$ of users completing a full course or maintaining use for X weeks.

- **Service Use:** e.g. X% reduction in mental-health-related GP visits or urgent care compared to baseline. (The Kooth model counted GP/medication reductions ³.)

- **Economic:** cost per user and cost-saving per user. Benchmark against analogues: Sleepio's cost was ~£45/year ¹; digital mental-health ROI often targets ~£200+ saving per person (as in Kooth). For addiction, compare: inpatient rehab ~£495/day ¹⁸ vs. app <£50/year.

A table of **per-patient cost comparisons** can illustrate this. For example:

Intervention	Typical Cost Per Patient	Key Outcomes (UK Context)
Inpatient addiction rehab	~£14,000 (28-day stay) ¹⁸	Very intensive support; long-term sobriety aim
Standard IAPT CBT therapy	~£300–£500 (6–8 sessions)	50% recovery rate target (NICE/IAPT goal)
Sleepio (digital CBT-i)	£45/year per user ¹	Clinically proven sleep improvement; NICE-recommended
Wysa (mental-health chatbot)	£5.90/year per user ²	Reductions in PHQ-9/GAD-7 scores; reduced referrals
OTOS platform (ADO+Addict)	TBD (pilot to define)	Expected improvements in ADHD symptoms, reduced substance use; <i>target cost</i> $\ll$ rehab cost

Cesaro above values highlight the potential savings: even a £100–200 digital intervention can look very attractive against thousands in alternative care. OTOS should gather baseline costs (e.g. average spend on adult ADHD treatment + addiction services per patient) and aim to undercut these through prevention/early intervention.

Preparing for Procurement and Scale-Up (12–18 Months)

If the pilot shows promise, OTOS should transition swiftly to procurement planning: - **Develop a Business Case:** Summarize pilot data (clinical improvement, cost per user, extrapolated savings) into an

NHS-style business case. Highlight alignment with CPICS strategy (e.g. ICS digital strategy mentions empowering people via online services ²¹). Use local data: note that Cambridgeshire has high rates of untreated alcohol dependency ¹⁹ and growing mental-health demand; emphasize OTOS's role in *prevention*.

- **Engage Procurement Early:** Connect with CPICS procurement officers and digital leads to discuss contracting. If pilot was successful, ask how to move to “trial to contract”. Propose contracting via frameworks like G-Cloud (if eligible as a SaaS), which ICSs commonly use ¹⁰. Alternatively, ICS might issue a local tender or add OTOS to an in-house digital formulary. Prepare all required docs (DTAC form, evidence packs, liability/clinical governance info) in advance.

- **Innovation Funding:** Keep watch for national funding: ICS-level transformation funds, NHS England's digital service funding, or inclusion in national programmes. For example, NHS England has ring-fenced transformation and innovation funding (recently recommended by the Health Tech Alliance) ¹¹. OTOS could seek small-scale funding (e.g. NHSX Test Beds or a CPICS Innovation Challenge) to bridge pilot-to-scale.

- **Foster Champions:** Identify clinician and patient champions who can advocate internally for the service. NHS adoption often depends on local “champions” (e.g. a lead psychiatrist or IAPT manager). Their support can ease the path from pilot to business-as-usual commissioning.

In practice, transitions to commissioned service often take 12–18 months: initial pilot (6–12 mo), followed by procurement (6 mo). OTOS should plan this timeline accordingly. The recent NHS focus on *value-based procurement* means OTOS's long-term outcome data will be vital – not just short-term pilot results.

Summary of Recommendations

- **Pilot Design:** Build an evidence-based pilot aligned with NICE/DTAC standards and ICS priorities. Include ADHD and addiction experts, use validated measures, and plan an economic analysis from Day 1.
- **Outcome Metrics:** Aim for clinically meaningful symptom reductions and demonstrable system impacts (fewer referrals, lower medication/hospital use). Benchmark against values reported in other digital tools (e.g. ~£200 savings per user ³, high engagement).
- **Cost-Benefit Framing:** Highlight how each OTOS user displaces a far costlier outcome (e.g. one inpatient relapse prevented equals ~£14,000 saved ¹⁸). Use local C&P data (e.g. ~4,500 dependent drinkers not in treatment ¹⁹) to show scale of need.
- **Procurement Path:** Plan to leverage NHS Innovation pathways (NIA, ICS innovation funds) during pilot, and move to ICS contract via standard frameworks afterwards. Engage procurement teams early and get DTAC-ready.
- **Strategic Alignment:** Emphasize to NHS that supporting ADHD as part of addiction care is a national priority (per the 2025 ADHD Taskforce ⁷). This aligns OTOS with the NHS's “prevention and digital” agenda.

By grounding the pilot in robust evidence standards and focusing on ROI-relevant outcomes, OTOS can make a compelling case to Cambridgeshire & Peterborough ICS. If the 12–18 month pilot demonstrates both improved patient outcomes and cost savings, the service can transition smoothly into a commissioned offering – joining the growing list of digital health tools adopted by the NHS ¹ ⁴.

Sources: NHS and NIHR guidelines and published evaluations (NICE DTAC/ESF, NHS England, NIHR, peer-reviewed and industry studies) of digital mental-health tools ⁵ ³ ² ¹ ⁴. Examples include evaluations and official guidance for Sleepio, Wysa, SilverCloud, Kooth and others ¹³ ¹⁷ ¹⁵ ¹⁸.

1 Resource impact summary report | Sleepio to treat insomnia and insomnia symptoms | Guidance | NICE

<https://www.nice.org.uk/guidance/mtg70/resources/resource-impact-summary-report-11074037437>

2 Wysa - AI-Powered Mental Health Support for Building Resilience - NHS Innovation Accelerator

<https://nhsaccelerator.com/innovations/wysa/>

3 An early economic evaluation of Kooth, a web-based mental health platform for children and young people with emerging mental health needs - PubMed

<https://pubmed.ncbi.nlm.nih.gov/38803649/>

4 Evidence-based smartphone app can help people reduce their alcohol intake - ARC

<https://arc-nenc.nihr.ac.uk/news/5847/>

5 6 Digital Technology Assessment Criteria (DTAC) - Key tools and information - NHS Transformation Directorate

<https://transform.england.nhs.uk/key-tools-and-info/digital-technology-assessment-criteria-dtac/>

7 20 NHS England » Report of the independent ADHD Taskforce: Part 1

<https://www.england.nhs.uk/long-read/report-of-the-independent-adhd-taskforce-part-1/>

8 The (digital) revolution must be scrutinised - why technology trials are vital to improve young people's mental health therapies | NIHR

<https://www.nihr.ac.uk/blog/digital-revolution-must-be-scrutinised-why-technology-trials-are-vital-improve-young-peoples-mental-health-therapies>

9 12 13 14 For Policy Makers

<https://www.bighealth.co.uk/for-policy-makers>

10 What are Integrated Care Systems? - 2022 Procurement Analysis

<https://www.tussell.com/insights/what-are-integrated-care-systems-2022-procurement-analysis>

11 How the NHS can Unlock Procurement Innovation | Procurement Magazine

<https://procurementmag.com/news/the-nhs-unlock-procurement-innovation>

15 Concern as a third of NHS mental health treatments shift online | Depression | The Guardian

<https://www.theguardian.com/society/2023/jul/02/concern-as-a-third-of-nhs-mental-health-treatments-shift-online>

16 Using Digital Mental Health Tools Shows Dramatic Cost Savings and Return on Investment for Mental and Behavioral Healthcare - MemorialCare Innovation Fund

<https://memorialcareinnovationfund.com/using-digital-mental-health-tools-shows-dramatic-cost-savings-and-return-on-investment-for-mental-and-behavioral-healthcare/>

17 Find an app for wellbeing and independence | GOV.WALES

<https://www.gov.wales/find-app-wellbeing-and-independence>

18 How Much Does Rehab Treatment Cost? | OK Rehab

<https://www.okrehab.org/rehab-questions/how-much-does-treatment-cost/>

19 Cambridgeshire & Peterborough Insight – Mental Health Needs Assessment – Population factors – Substance Use – Alcohol use

<https://cambridgeshireinsight.org.uk/mentalhealthneedsassessment/mental-health-needs-assessment-chapter-two/substance-use/alcohol-use/>

21 Digital Strategy | CPICS Website

<https://www.cpics.org.uk/digital-strategy/>